

Vantage: Using Dynamic Difficulty Adjustment to Pace Emotional Intensity in Serious Games

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Abstract. Dynamic Difficulty Adjustment (DDA) is commonly used to regulate challenge by adapting game parameters in response to player performance. In serious games focused on stress, cognition, or emotional well-being, however, adaptation may need to support continued engagement during difficult moments rather than optimize performance alone. This paper presents *Vantage*, a 2D serious game prototype that reframes DDA as emotion-aware task pacing. In the game, negative thoughts are externalized as hostile entities, while positive thoughts and power-ups function as supportive resources. *Vantage* monitors gameplay-derived indicators associated with interaction strain and computes a score intended to guide environmental intensity and the introduction of temporary stabilization periods. We argue that this framing extends DDA beyond challenge balancing and offers a useful design direction for serious and mental health games.

Keywords: dynamic difficulty adjustment, serious games, mental health games, affective adaptation, emotional regulation, cognitive reframing

1 Introduction

Dynamic Difficulty Adjustment (DDA) is a game design strategy in which game parameters change in response to player performance during play. Traditionally, DDA has been used to keep challenge within a target range by adapting factors such as enemy behavior, speed, health, or resource availability [1]. In serious games, this approach is often framed as a way to reduce boredom when tasks are too easy and frustration when they become too difficult [14]. Prior works have refined DDA through interface support, combat variables, task presentation, AI agents, and physiological sensing, often to guide adaptation and evaluate flow-oriented experiences [10, 15, 17, 18].

Serious game research occasionally adopts DDA because these systems are expected to both engage players and support a broader external goal [20]. Yet adaptive mechanics in this space are still often evaluated through enjoyment,

task completion, or performance outcomes [4]. Recent works have questioned this heavy reliance on flow and argued for multimodal approaches that adapt difficulty according to the broader goals of the game rather than flow alone [3, 4]. This distinction is especially important in serious games for mental health, where adaptation may need to support a player’s ability to remain with difficult material in a manageable way rather than simply maintain enjoyment or performance [3, 4, 18].

This work explores that design space through our serious game, *Vantage*. *Vantage* is a 2D game that engages thought patterns associated with student stress through a thought-reframing exercise. Drawing from Cognitive Behavioral Therapy (CBT)-oriented thinking, the game externalizes thoughts as visible elements in play so that players can confront and interpret them through action [11]. Players shoot hostile skull-like enemies that reveal positive or negative thought elements when destroyed, while supportive power-ups help them continue engaging with the task. Rather than using DDA only to maintain challenge or enjoyment, *Vantage* adapts the pace of the game when gameplay behavior suggests that the player is under strain.

This paper makes three contributions. First, it reframes DDA in serious games as a mechanism for emotion-aware task pacing rather than only balancing challenge for enjoyment or performance. Second, it presents the design of *Vantage* as a working mini-game prototype that aligns adaptive pacing with the broader goals of a mental health-oriented serious game. Third, it argues that multimodal measures of challenge in mental health serious games may support continued engagement during cognitively or emotionally demanding interaction.

2 Related Work

2.1 Dynamic Difficulty Adjustment in Games

Dynamic Difficulty Adjustment (DDA) emerged as an alternative to static difficulty selection, allowing games to adapt challenge during play in response to player performance [1, 2]. Early work largely framed DDA as a way to maintain a balance between challenge and skill by adjusting factors such as enemy behavior, resource availability, or pacing. In this view, adaptation was used to prevent gameplay from becoming either too easy or too frustrating, and this framing became closely associated with flow-oriented interpretations of play [5].

More recent work has questioned whether flow should remain the dominant lens for understanding DDA. Guo et al. argue that DDA research has become overly dependent on flow as a universal design goal, limiting how adaptive systems are imagined and evaluated [3]. Rather than assuming that performance equilibrium is always the desired outcome, they suggest that DDA should be understood in relation to the broader goals of a game and the specific forms of difficulty it is trying to regulate [3]. This critique is especially important for serious games for mental health, where balancing challenge and skill may not be the only adaptive objective.

2.2 DDA in Serious Games and Training Contexts

DDA has also been applied in serious games, simulation training, and instructional contexts. A systematic review by Seyderhelm and Blackmore found that adaptive systems in serious games most often rely on performance-based measures, while a smaller subset incorporate physiological or cognitive indicators [4]. Their review further suggests that approaches using multiple measures may better support learning and training outcomes, especially when adaptation is aligned with pedagogical goals rather than gameplay performance alone [4].

At the same time, this review demonstrates that adaptation in serious games is still most often discussed in terms of improving flow-aligned metrics rather than supporting how players move through difficult material [3, 4, 18]. Even when cognitive load or affective state is considered, the adaptive goal is usually to improve mastery, maintain involvement, or optimize task success [3, 8]. This leaves an important gap for wellness-oriented and mental health serious games, where the design problem may be less about maximizing success and more about helping players remain engaged with emotionally or cognitively demanding tasks in a manageable way.

2.3 Affective and Physiology-Based Adaptation

Research in affective computing has expanded DDA by incorporating signals such as facial expression and physiological arousal to estimate player state and adjust gameplay accordingly [6, 7, 8, 9]. These approaches move beyond simple performance metrics by attempting to account for how a player may be feeling during play, not only how well they are performing. Prior works have used anxiety-related physiological feedback, self-reported emotion, and facial expression analysis to guide dynamic adaptation [7, 8, 9].

However, these examples of affect-aware DDA remained focused on preserving flow-oriented balance and did not combine subjective and objective adaptation for the goals of the game [3]. In these serious-game contexts, physiological and affective adaptation have been used to support performance for rehabilitation or training outcomes, but DDA does not shape the pace of a difficult task itself to mitigate strain [3, 4]. *Vantage* builds on this literature but shifts the purpose of adaptation. Rather than using external sensing hardware, it relies on gameplay-derived behavioral signals to estimate interaction strain, and it uses that estimate to reduce pace and intensity when the player appears overwhelmed. In this sense, *Vantage* extends DDA toward emotion-aware task pacing in serious games.

3 System Design

3.1 Core Interaction Loop

Vantage is a 2D action-based serious game that externalizes stress-related cognition through a dynamic combat environment (See Figure 1). Players control

an on-screen avatar and use a bow to defend against incoming skull-like enemies that move down the screen while firing at the player. These hostile entities symbolically represent intrusive or negative thought patterns [8, 11]. The core interaction loop centers on confronting and managing waves of enemies while navigating rising pressure in the play space. Figure 1 shows the primary interaction flow of *Vantage*, from difficulty selection and symbolic combat to reflective feedback at the end of play.



Fig. 1: *Vantage* gameplay flow. Players begin by selecting a difficulty, engage hostile thought representations during play, and receive reflective feedback through reframed and defeated thoughts at the end of the session.

When enemies are destroyed, they reveal visible thought elements categorized as either negative or positive, as shown in Table 1. Drawing from CBT [11], the negative thought patterns were adapted from the Automatic Thoughts Questionnaire (ATQ) [12] and refined to fit the context of the game. These negative thoughts appear as visible elements that float downward when enemies are destroyed, allowing players to encounter representations of maladaptive cognition patterns during play. Positive thought patterns function as supportive counterparts and were informed by the ATQ-P [13] questionnaire and adapted. Resources tied to positive thoughts can be collected to provide temporary gameplay advantages, such as increased firepower, a protective shield, or a slowdown effect. In this way, the play loop connects active engagement with symbolic thought management and support. Together, these paired thought structures are intended to represent cognitive patterns that therapeutic practice often seeks either to reframe or to reinforce.

3.2 Behavioral Indicators of Interaction Strain

Rather than attempting to detect emotion directly from the player, *Vantage* monitors responses from gameplay that may suggest interaction strain [8]. The system evaluates five indicators: hit frequency, combat accuracy, survival inter-

Table 1: Examples of negative thought patterns and supportive responses in *Vantage*.

Negative thought	Supportive response	Gameplay function
“Nothing I do matters.”	“You acted anyway. The Archive recorded the outcome.”	Reframes hopelessness by emphasizing that action still has meaning.
“If I slow down, I fall behind.”	“You slowed down and still accomplished your task.”	Challenges urgency by linking slower pacing with continued progress.
“Everyone else is stronger than me.”	“Strength was not required. Persistence was.”	Shifts focus from comparison to endurance and steady effort.
“I should have started sooner.”	“You started when you could. That was enough.”	Counters self-blame by validating meaningful action in the present.

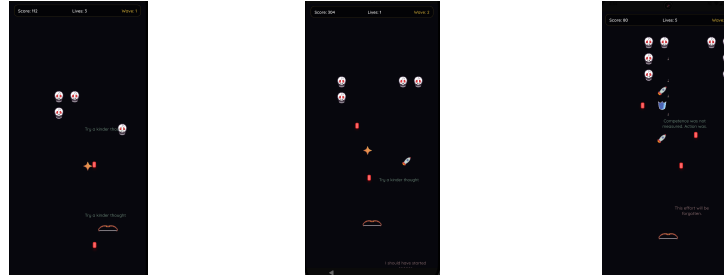
val, time since the last shot, and time since the last enemy was destroyed. Together, these measures aim to capture both immediate pressure and reductions in activity (See Equation 1).

A sustained pattern of frequent hits, reduced accuracy, shorter survival intervals, and longer gaps between actions is used as a basis for estimating when the pace of the task may be contributing to interaction strain. This estimate is intentionally modest to allow for iteration across contexts. It is not treated as a measure of player distress but as a gameplay-based proxy for possible strain within the immediate interaction.

3.3 Emotion-Aware Task Pacing

When the system detects sustained signs of interaction strain, *Vantage* reduces environmental intensity in an effort to create a more manageable pace. This is done by slowing enemy formation speed, reducing enemy movement speed, and increasing the likelihood of beneficial drops. Together, these changes are intended to create a stabilization window in which the player has more room to recover tempo, respond to supportive resources, and continue interacting with the task.

As the player’s behavior returns to a more stable range, the game gradually restores intensity rather than shifting difficulty abruptly. This approach differs from DDA systems that primarily aim to preserve challenge-skill balance alone [1, 2, 5]. In *Vantage*, adaptation is used to pace a cognitively and emotionally demanding interaction in a way intended to help the player remain engaged without abrupt escalation. Figure 2 shows how thought statements and supportive resources appear directly in the combat space, reinforcing the game’s use of symbolic play to represent stress-related cognition.



(a) Thought cue during play. (b) Supportive resource appears. (c) Multiple thought and support elements.

Fig. 2: Thought patterns and supportive resources in *Vantage*. During play, hostile enemies are accompanied by visible thought statements, while supportive items such as power-ups appear in the same space to help the player continue engaging with the task.

3.4 Adaptive Difficulty Model

The adaptive controller in *Vantage* uses a multi-factor pressure model to estimate when the pace of the task may be exceeding the player's ability to sustain engagement. Rather than relying on a single success or failure measure, the model combines five normalized behavioral metrics: hit frequency, combat accuracy, survival interval, shooting inactivity, and kill inactivity. These indicators are weighted to reflect both immediate danger and reduced interaction tempo. Hit frequency and accuracy pressure each contribute 25% of the total estimate, kill activity contributes 20%, and survival interval and shooting inactivity each contribute 15%.

These indicators are combined into a bounded pressure score:

$$\text{Pressure} = \sum_{i=1}^5 w_i m_i \quad (1)$$

where m_i represents each normalized behavioral metric and w_i its corresponding weight.

To smooth adaptation over time, the system maintains an emotional intensity value using an exponential moving average:

$$EI_t = \alpha \cdot \text{TargetEI}_t + (1 - \alpha) \cdot EI_{t-1} \quad (2)$$

Here, EI_t represents the smoothed intensity value used to update gameplay parameters over time. The model also includes a decay mechanism: when the player avoids taking damage for several seconds, the pressure estimate gradually decreases rather than remaining artificially elevated. This helps prevent a brief spike in difficulty from causing an extended reduction in challenge after the player has already recovered.

Difficulty adaptation is governed by threshold ranges derived from this pressure estimate. When pressure rises above 0.7, the system enters a recovery-oriented mode and reduces effective intensity. When pressure falls below 0.3, challenge is allowed to increase gradually.

Under higher strain, the system reduces intensity more aggressively than it restores it during recovery. This asymmetry is intentional. Because the game is designed around stressful and supportive thought representations, we argue that it is more important to create space for reorientation than to maximize escalation. The resulting pressure estimate modulates three gameplay parameters in real time: enemy formation speed, enemy movement speed, and beneficial drop probability. As strain increases, hostile buildup slows and supportive resources become more accessible. Together, these design methods are intended to re-balance the environment in ways that may help players continue engaging with difficult material without being pushed too quickly by the pace of the task.

3.5 Limitations and Future Work

Vantage is presented as a design-oriented prototype and does not claim that its gameplay-derived estimate of interaction strain directly measures emotional state. Instead, the adaptive model uses behavioral signals as proxies for possible strain during play. While this approach is useful for responsive game design, it should not be interpreted as a clinical measure of overwhelm or distress. Additionally, this paper does not include comparative evaluation against non-adaptive versions of the game or prior DDA approaches. An important next step is to conduct controlled studies that examine whether emotion-aware task pacing changes how players experience pressure, recovery, and engagement under stress.

4 Conclusion

This paper presented *Vantage*, a serious game prototype that uses dynamic difficulty adjustment for emotion-aware task pacing rather than optimizing performance or enjoyment alone. By externalizing negative and positive thoughts within a symbolic combat environment and adapting task pace in response to a gameplay-derived estimate of interaction strain, *Vantage* illustrates how DDA might be refined to better support players during demanding moments of play. We argue that this framing offers a promising direction for serious and mental health games, where adaptation may do more than balance challenge by also helping players remain engaged with difficult yet meaningful interactions during overwhelming moments.

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